MIL-I-12597C (PA)
23 May 1975
Superseding
MIL-I-12597B (MU)
21 August 1969

MILITARY SPECIFICATION

IGNITER, BLASTING FUSE, FRICTION TYPE, M3A1 LOADING, ASSEMBLING AND PACKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading assembling and packing for one type of initiating device designated as Igniter, Blasting Fuse, Friction Type, M3A1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-Z-325 - Zinc Plating (Electrodeposited)
QQ-P-416 - Plating, Cadmium (Electrodeposited)

MILITARY

*MIL-P-116 - Preservation, Methods of
*MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For
MIL-F-20412 - Fuse, Safety (Bickford Type)

FSC: 1375

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables For Inspection by Attributes (ABC-STD-105)
MIL-STD-1234 - Pyrotechnics, Sampling, Inspection and Testing

*(For meaning of asterisk (*) see paragraph 6.4)

DRAWINGS

U.S. ARMY PICATINNY ARSENAL

8833721 - Igniter, Blasting Fuse, Friction Type, M3A1, Assembly
8835011 - Box, Packing, Ammunition for Igniter, Blasting Fuse, Friction Type, M3A1
8835012 - Carton, Packing, Ammunition for Igniter, Blasting Fuse, Friction Type, M3A1

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

*2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Designation E300 - Industrial Chemicals, Recommended Practice for Sampling

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Penn. 19103).

3. REQUIREMENTS

3.1 Material. - Materials shall be in accordance with applicable drawings and specifications.

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3.2 Igniter. - The igniter shall comply with all requirements specified on Drawing (dwg.) 883721 and with all requirements specified in applicable specifications.

3.3 The igniter shall not ignite when subjected to the tumbling test.

3.4 Moisture content of paper products. - The moisture content of the tube, sleeve and paper packing components shall not exceed six (6) percent at the loading station at the time of loading and packing the assemblies.

3.5 Functioning. - The igniter blasting fuse assembly shall ignite the safety fuse when tested.

3.6 Quick leak test. - The packed carton sealed in the barrier bag shall show no evidence of air or water leakage when tested.

3.7 Workmanship. - All parts and assemblies shall be fabricated and loaded in a thorough workmanlike manner, and all manufacturing, processing, and assembly operations shall be correctly performed. They shall be clean and free of burrs, sharp edges, unblended radii, surface defects, chips, dirt, grease and oil (except where specifically required), corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part or assembly nor shall they be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. Required packing shall be dry.

3.8 First Article Inspection. - This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions. - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. - The following types of inspection shall be conducted on this item.

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a. First Article Inspection

b. Quality Conformance Inspection

4.3 First Article Inspection

4.3.1 Submission. - The contractor shall submit a first article sample (see 6.1b) as designated by the Contracting Officer. The first article sample shall consist of twenty-five (25) complete sets of parts and subassemblies (i.e. 25 each of every component and subassembly) and fifty (50) loaded igniter assemblies. The first article shall be evaluated in accordance with the provisions of 4.3.2 and 4.3.3.

4.3.2 Inspections to be performed. - See MIL-A-48078 and paragraphs 4.4.2.1 through 4.4.2.9 and Table I specified herein.

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TABLE I

FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-I-12597C (PA)

PARAGRAPH *	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Igniter, Blasting Fuse, Friction Type M3A1, Parts and Assemblies				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Parts				
	Dwg. 8833721				
	Examination for defects	25(a)		3.2	4.4.2.1 through 4.4.2.6
	Subassemblies				
	Dwgs. 8833719 and 8833722				
	Examination for defects	25(a)		3.2	4.4.2.7 and 4.4.2.8
	Assemblies				
	Dwg. 8833721				
	Examination for defects	50(b)		3.2	4.4.2.9
	Testing				
	Functioning	50		3.2	4.5.4
	Salt Spray (Ferrule & Clip)	10(c)		3.2	4.5.6
NOTES:	(a) 25 each and every component and subassembly				
	(b) Inspected prior to functioning				
	(c) Ten each, randomly selected from parts	(a)			

NOTES:

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4.3.3 Rejection. - (See MIL-A-48078)

4.4 Quality conformance inspection

*4.4.1 Inspection lot formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of igniters shall contain:

a. Parts of one kind from one manufacturer.

b. Composition ingredients from not more than one supplier.

c. Igniter and friction composition of one or more batches produced under one continuous set of operating conditions, with each batch of the composition having been subjected to the same physical mixing process intended to make the batch homogeneous.

4.4.2 Examination. - (See MIL-A-48078). Unless otherwise specified in the classification of defects and test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.4.4.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-I-12597C (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
*4.4.2.1	Tube		8833712
			NEXT HIGHER ASSEMBLY 8833722

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131	Length		0.40%	3.2	Scale
132	Outside diameter		0.40%	3.2	Gage
133	Inside diameter		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-I-12597C (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
*4.4.2.2	Clip, Safety		8833713
			NEXT HIGHER ASSEMBLY 8833721

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131	Length		0.40%	3.2	Scale
132	Thickness of wire		0.40%	3.2	Gage
Minor					
201	Diameter of eyelet		0.65%	3.2	Scale
202	Radius of large bend		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-I-12597C (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
#4.4.2.3	Ferrule		8833714
			NEXT HIGHER ASSEMBLY
			8833722

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131	Diameter		0.40%	3.2	Gage
132	Thickness		0.40%	3.2	Gage
Minor					
201	Width between prongs		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-I-12597C (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
*4.4.2.4	Cup, Igniter		8833716
			NEXT HIGHER ASSEMBLY
			8833719

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131	Thickness		0.40%	3.2	Gage
132	Diameter of open end		0.40%	3.2	Gage
133	Diameter under flange		0.40%	3.2	Gage
Minor					
201	Pierced hole missing		0.65%	3.2	Visual
202	Shellac coating missing		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-I-12597C (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.5	Sleeve				8833717
					NEXT HIGHER ASSEMBLY 8833721
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131	Length		0.40%	3.2	Scale
132	Location of hole		0.40%	3.2	Gage
Minor					
201	Outside diameter		0.65%	3.2	Gage
202	Diameter of small hole		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-I-12597C (PA)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
*4.4.2.6	Cap	1 OF 1	8833718
			NEXT HIGHER ASSEMBLY 8833722

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131	Inside diameter		0.40%	3.2	Gage
132	Location of large hole		0.40%	3.2	Gage
Minor					
201	Length		0.65%	3.2	Scale
202	Outside diameter		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-I-12597C (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8833719
*4.4.2.7	Assembly, Igniter Charge Wire				NEXT HIGHER ASSEMBLY 8833722
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Friction composition coating exceeds 5 spaces on friction wire		100.0%	3.2	Visual
2	Charge loose or not consolidated in cup		100.0%	3.2	Visual
3	Binder coating missing		100.0%	3.2	Visual
Major A					
101	Friction composition coating less than 3 spaces on friction wire		0.40%	3.2	Visual
102	Igniter charge composition within lip area		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-I-12597C (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
*4.4.2.8	Igniter Charge Cap Assembly		8833722
			NEXT HIGHER ASSEMBLY 8833721

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Staple missing		0.40%	3.2	Visual
102	Pull cord insecurely tied to igniter charge wire and to cap		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-I-12597C (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
*4.4.2.9	Igniter, Blasting Fuse, Assembly		8833721
			NEXT HIGHER ASSEMBLY 8833721

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Safety clip missing, not fully engaged or incorrectly located		100.0%	3.2	Visual/Manual
Major A					
101	Assembly damaged		0.40%	3.2	Visual
102	Components, other than safety clip, missing		0.40%	3.2	Visual
103	Tube loose		0.40%	3.2	Manual
Minor					
201	Length, maximum		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-I-12597C (PA)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
*4.4.2.10	Carton	1 OF 1	8835012
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131	Carton damaged to extent that contents are exposed		0.40%	3.2	Visual
132	Sealing incomplete or inadequate		0.40%	3.2	Visual
133	Carton, filler or paper products .. obviously damp or wet		0.40%	3.2	Visual
Minor					
201	Contents loose		0.65%	3.2	Manual
202	Marking misleading or unidentifiable		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-I-12597C (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8835012
*4.4.2.11	Sealed Packed Barrier Bag Prior to Packing into Box				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Waterproofness of barrier bag destroyed by rupture, puncture or separation of heat seal		0.40%	3.2	Visual
102	Evidence of crushed box		0.40%	3.2	Visual
NOTE:	Package should be opened if there is evidence of crushing.				
Minor					
201	Heat seal of barrier bag located so that one or more subsequent seals cannot be applied		0.65%	3.2	Visual
202	Marking of bag misleading or unidentifiable		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-I-12597C (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
#4.4.2.12	Sealed Packing Box		8835011
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Box damaged to extent that contents are exposed		0.40%	3.2	Visual
102	Hardware broken or loose		0.40%	3.2	Visual/Manual
103	Strapping missing, broken or loose		0.40%	3.2	Visual/Manual
104	DOD symbol misleading or unidentifiable		0.40%	3.2	Visual
Minor					
201	Hardware improperly engaged		0.65%	3.2	Visual
202	Strapping improperly assembled		0.65%	3.2	Visual
203	Metallic seal unsealed or incorrectly positioned		0.65%	3.2	Visual
204	Marking misleading or unidentifiable		0.65%	3.2	Visual
205	Contents loose		0.65%	3.2	Manual
206	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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4.4.3 Testing

4.4.3.1 Chemical Sampling. - A representative sample shall be randomly selected for the applicable tests specified in 4.4.3. The procedure described in ASTM E300 shall be used to withdraw the samples. The applicable tests specified in 4.5 shall be performed using prescribed analytical procedure for replicate determination given in standard analytical textbooks.

4.4.3.2 Moisture content. - (See Table II) Major A defect.

TABLE II

Material	
Paper packing components	(See dwg. 8835012)
Tube	(See dwg. 8833712)
Sleeve	(See dwg. 8833717)

The contractor shall provide adequate controls to insure compliance with the requirements of 3.4 and shall test for verification at least one sample of each material from each eight hours production of igniters. Composite samples shall not be used. If the moisture content of a sample exceeds the requirement, and loading has not begun, that quantity of material represented by the sample shall be rejected. If assemblies have been loaded with material containing excessive moisture, the remaining material that is represented by the sample together with all assemblies loaded with the non-conforming material shall be rejected. The test shall be performed as specified in 4.5.1.

4.4.3.3 Determination of igniter and friction composition

4.4.3.3.1 Igniter composition (see dwg. 8833719) Major A defect. - A sample of three igniter blasting fuse assemblies shall be selected from each quantity or sublot as representative of each batch or mixture of composition. Test shall be performed as specified in 4.5.2. If the igniter composition fails to meet the specified requirements, the quantity or lot shall be rejected.

4.4.3.4 Tumbling. - Critical defect - This test shall be conducted 100 percent on all igniter blasting fuse assemblies prior to assembly of ferrule. The test shall be performed as specified in 4.5.3.

4.4.3.5 Functioning. - Major A defect

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4.4.3.5.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the requirements specified, two hundred (200) igniter blasting fuse assemblies shall be selected from each lot for test. The lot shall be rejected if six or more assemblies fail to ignite the safety fuse. The test shall be performed as specified in 4.5.4.

4.4.3.5.2 After three consecutive lots have met the acceptance criteria in 4.4.3.5.1, eighty (80) igniter blasting fuse assemblies shall be selected from each lot for this test. The lot shall be rejected if three or more assemblies fail to ignite the safety fuse.

4.4.3.6 Quick leak test of sealed packed barrier bag. - Major A defect - Ten (10) cartons in sealed barrier bags shall be selected from each lot for this test. If two (2) or more packages show evidence of air leakage during test or water leakage when package is opened the lot shall be rejected. The test shall be performed as specified in 4.5.5.

4.4.3.7 Salt spray test. - (see Table III) Major A defect.

TABLE III

Parts For Test

Clip, safety (see dwg. 8833713)
Ferrule (see dwg. 8833715)

Sampling plan for this test shall be in accordance with applicable protective finish specified on drawing. Test shall be performed as specified in 4.5.6.

4.4.4 Inspection Equipment. - The inspection equipment required to perform the inspections and tests prescribed in this specification is identified in the 'Paragraph Reference/Inspection Method' column in the tables starting with paragraph 4.4.2.1 and the test method paragraphs (see 4.5). The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078, and section 6.2 herein.

4.5 Test methods and procedures

4.5.1 Moisture content of paper products. - The moisture content shall be determined with Method 102.1 of Standard MIL-STD-

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1234 except that a 10 gram sample of each paper item shall be used and dried for 2 hours.

4.5.2 Determination of igniter composition

4.5.2.1 Charcoal. - Place three loaded igniter cups in a 30 milliliter beaker which has been dried at 100 degrees Centigrade (C.) plus or minus 5 degrees C for one hour; cool in a desiccator, and weigh. Add to the beaker 20 milliliters of water. Heat the contents of the beaker to boiling and boil until the composition is completely leached from the igniter cups. Filter the contents of the beaker, using a tared Selas filtering crucible Number 3010 or equivalent. Transfer the residue to the crucible quantitatively, leaving the cleaned igniter cups in the beaker. Wash the residue four times with hot water. Dry the crucible for one hour at 100 degrees C. to 105 degrees C., cool in a desiccator and weigh. Retain the filtrate for the determination of potassium chlorate and dextrin. Consider the gain in weight of the crucible as weight of charcoal in the sample. Dry the beaker and leached igniter cups at 100 degrees C. plus or minus 5 degrees C. for one-half hour. Consider the loss in weight as the weight of the dry sample. Calculate the gain in weight of the crucible as percentage of charcoal in the dry sample.

4.5.2.2 Potassium chlorate. - Dilute the filtrate obtained in 4.5.2.1 to 250 milliliters in an accurately calibrated volumetric flask. Transfer a 25 milliliter aliquot with a calibrated pipette to a 500 milliliter Erlenmeyer flask and dilute to approximately 150 milliliters. Add approximately 50 milliliters of 18N sulfuric acid. Stopper the flask with a rubber stopper equipped with a Bunsen valve and heat to almost the boiling point. Add 50 milliliters of a standardized solution of approximately 4 percent ferrous ammonium sulfate (Mohrs Salt) by means of a calibrated pipette and boil the solution for 2 minutes. Cool the solution, loosen the valve and titrate the excess Mohrs salt with a standard 0.1N solution of potassium permanganate. Concurrently, conduct a blank determination using the same quantities of the same reagents used for the analysis of the sample. Calculate the percentage of potassium chlorate in the sample as follows:

$$\text{Percentage of potassium chlorate} = \frac{20.43\ (V-v)N}{W}$$

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where:

V = volume of potassium permanganate required for blank, titration, ml.

v = volume of potassium permanganate required for sample titration, ml.

N = normality of potassium permanganate solution

W = weight of the sample, gms.

4.5.2.3 Dextrin. - Calculate the percentage of dextrin in the sample by subtracting from 100 the combined percentage of charcoal and potassium chlorate.

4.5.3 Tumbling. - The igniter blasting fuses, without ferrules, shall be subjected to the tumbling test for a minimum of 20 plus or minus two (2) minutes. Approximately 500 igniters shall be tested simultaneously. Following test, 100 percent examination shall be made to segregate those igniters which ignited during tumbling. The tumbler box shall be made of 3/4 inch plywood with inside dimensions 36 inches long by 36 inches wide by 24 inches deep, and with four (4) baffles 9 inches wide by 18 inches deep centered in and perpendicular to the four (4) walls. The box with bottom canted at 72 plus or minus two (2) degrees from the horizontal shall be rotated at nine (9) plus or minus one (1) revolutions per minute.

4.5.4 Functioning. - A piece of safety fuse, complying with specification MIL-F-20412, 3/16 inches by 3 inches shall be inserted not less than 3/4 of an inch into the igniter until firmly held in place by the ferrule prongs. The igniter, with the safety fuse in place, shall be assembled to the applicable test fixture specified in 4.4.4, and the igniter shall be functioned by pulling the friction wire. Any igniter which fails to ignite the safety fuse shall be classed defective.

4.5.5 Quick leak test. - (Vacuum Differential Method) - The filled and sealed packages, after conditioning at ambient temperature (70 plus or minus 5 degrees F.) for at least four (4) hours, shall be tested for leakage by submerging in water contained in a vacuum vessel. A vacuum differential of 216 mm. (8 1/2 inches) of mercury from ambient conditions shall be drawn and held for 30 seconds min. while observing for leakage of air. A steady stream or recurring succession of bubbles from any surface or seam shall be cause for rejection. Where there is no emission of bubbles observed, the package shall be dried, opened and inspected. Evidence of water within the barrier, or on the carton shall also be cause for rejection.

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4.5.6 Salt spray test. - This test shall be performed in accordance with the Specification QQ-Z-325 or QQ-P-416 whichever is applicable. The duration of the test shall be 48 hours in lieu of 96 hours.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A. - The igniter blasting fuse assemblies shall be preserved and packaged in accordance with dwg. 8835012.

5.2 Packing

5.2.1 Level A. - The igniter blasting fuse assemblies shall be packed in accordance with dwg. 8835011.

5.3 Marking. - The cartons and boxes shall be marked as specified on dwgs. 8835012 and 8835011.

6. NOTES

6.1 Ordering Data. - Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Provisions for submission of first article samples.

c. Provisions of MIL-A-48078.

6.2 Submission of Inspection Equipment Designs for Approval. - (See MIL-A-48078) - Submit equipment designs as required, to Commander: Picatinny Arsenal, ATTN: SARPA-QA-T-T, Dover, New Jersey 07801.

6.3 Distribution of ammunition data cards. - Distribution of data cards shall include the following: Commander: Picatinny Arsenal, ATTN: SARPA-QA-A-M, Dover, New Jersey 07801.

6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletion) from the previous issue were made. This

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was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: FSC-1375-A-141

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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